

MOHAMMED GHALIB

AI Engineer • RAG & Agentic Systems • LLM Applications in Production

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PROFESSIONAL SUMMARY

AI engineer who builds LLM systems that are reliable enough to ship. I design retrieval and agentic pipelines that plan their own lookups, ground every answer in real sources, verify their own output, and refuse to guess when they are not sure. My focus is the hard part of applied AI: making a model trustworthy in front of real users, and proving it with evaluation rather than opinion.

I lead Sheria AI at Datawise Africa, a legal research platform for Kenyan law that cites, quotes, and explains the actual provision behind every answer. It runs an agentic LangGraph pipeline: triage, a research step that plans its own tool lookups over case-law and constitutional corpora, then a draft, verify, and repair loop, backed by triage and blind side-by-side evaluation harnesses. I built the same class of system at Ummly, an Islamic knowledge platform, on Pinecone RAG with reranking, a confidence floor, query rewriting, and bilingual answers with per-excerpt citations.

I build the whole system around the model, not just the prompt. I write the Django and FastAPI backends, the React and React Native apps, and I containerise, deploy, and maintain them in production, so I can take an AI feature from idea to a live, monitored service.

PROFESSIONAL EXPERIENCE

AI & Software Engineer, Datawise Africa

Feb 2026 – Present • Nairobi / Remote

Kenyan software and AI company. I lead Sheria AI end to end and build in-house AI systems with the team.

- Lead **Sheria AI** ([sheria.africa](#)), a legal research platform for Kenyan law. Handed to me as a project to lead, and delivered as a working platform now in production. It takes a question in plain language and answers it by citing, quoting, and explaining the actual provision from the Constitution, statutes, and decided cases, so advocates and law students find the authority in minutes instead of an afternoon of reading.
- **Rebuilt the core** from a single retrieve-then-answer call into an **agentic pipeline on LangGraph**: each question is triaged, then researched by a model that plans its own lookups against case-law and Constitution tools, and each answer is drafted, verified, and repaired before the user sees it, with a coverage check so no part of a multi-part question is dropped.
- **Engineered the retrieval layer** for a domain where a near-miss is a wrong answer. Chunked the case index so search reads the court's actual reasoning rather than the header, tuned the relevance thresholds that keep unrelated matters out, added exact-name case lookup, and enforced cited sources with a firm no-fabrication rule and a plain refusal when nothing relevant exists.
- **Route work across model tiers**: a reasoning model for triage and tool-planning, a separate model chosen by blind A/B for the user-facing writing, and a lightweight fallback that keeps a turn alive when a tier fails.
- **Built the evaluation harnesses** the team ships on: a triage scorecard, and a blind side-by-side comparison of candidate answer models, so prompt and model changes go out on evidence. The judgment lives in editable prompt files, so behaviour is tuned in prose rather than code.
- **Built the streaming chat UI** (React Router SSR, TypeScript) that renders answers token by token over server-sent events, with ranked source cards, and deliver in-house projects across the academic and health sectors as team work on Git and GitHub, with pull-request review, CI, a quality gate, and versioned releases.

Software Engineer, Digital Appex

Dec 2024 – Jan 2026 • Remote

Kenyan technology company building its own products and delivering projects for clients. I took on incoming client work and maintained the company's internal systems.

- **Built AI solutions for clients end to end**, from the first requirements conversation through to live deployment and handover: chatbots, retrieval-based (RAG) assistants, internal business systems, and operational dashboards, for organisations including healthcare and education providers.
- **Built and maintained the company platform** ([digitalappex.co.ke](#)), a Django and DRF backend with a React frontend, including its AI chatbot, and kept it live in production with payments, role-based access, background jobs on Celery and Redis, Docker, and GitHub Actions CI on a DigitalOcean server.

Software Engineer, Bee MultiScent Ltd

Feb 2023 – Dec 2024 • Hybrid

Scent-technology company. I ran the AI diffuser work, the internal platform, and the website.

- **Developed the AI for a smart scent diffuser**, automating how and when the company's IoT diffuser released scent.
- **Built the company portal** (Django and DRF backend, React frontend, real-time dashboards, WebSocket notifications, and a companion React Native app on the same API) and the company website.

SELECTED AI PROJECTS

Independent and contract work delivered alongside the roles above.

- **Ummly (formerly Islamify), Islamic knowledge platform** (ummlly.co.ke). Built solo a **Pinecone RAG** assistant grounded in the Qur'an and Hadith: server-side embeddings, **bge-reranker reranking**, a confidence floor that returns a low-confidence message instead of guessing, intent classification, query rewriting, bilingual EN/AR answers, and per-excerpt citations. LLaMA via Groq with a Google Gemini fallback. Shipped the full product around it: Django and DRF backend, React Router (SSR) web app, and an Expo / React Native mobile app, containerised and deployed via GitHub Actions to a DigitalOcean server.
- **Non-AI full-stack platforms for clients:** Collaborative Approach, a referral and tour-booking SaaS with M-Pesa, PayPal, and Stripe payments, and a multi-portal school management system, both on React, Django, and PostgreSQL with CI/CD.

CORE SKILLS

AI Ownership: Taking an AI product from problem statement to live service: architecture, retrieval design, evaluation strategy, deployment, and the trade-off calls in between

Reliability Judgment: Knowing where a model will fail and proving it before users do: grounding, refusal behaviour, coverage checks, and evidence-backed prompt and model changes

Ways of Working: Agile/Scrum, Git and GitHub, branching and pull-request review, Jira, Slack, remote and hybrid teams

TECHNICAL SKILLS

AI & LLM Engineering: Agentic pipelines on LangGraph (plan, research, verify, repair), tool-calling / function-calling, RAG design, prompt engineering, intent routing, hallucination control, grounding & source citation, graceful refusal on low confidence, evaluation harnesses (triage scorecards, blind side-by-side A/B)

LLMs & Providers: OpenAI, LLaMA (via Groq), Google Gemini, DeepSeek, reasoning models (gpt-oss), multi-model routing, tiering, and fallback orchestration

Retrieval & Vectors: Pinecone, ChromaDB, Sentence-Transformers embeddings, chunked indexing, reranking (bge-reranker), similarity-threshold retrieval, hybrid lookup (vector search + exact match), dual-corpus retrieval

AI Frameworks & Serving: LangGraph, LangChain, token streaming over server-sent events (SSE), Celery for async tasks, prompt-as-config (behaviour tuned in editable prompt files)

Backend & APIs: Python, Django, Django REST Framework, FastAPI, REST, WebSockets/Channels, JWT, PostgreSQL, Redis, SQLite

Frontend: React, React Router (SSR), TypeScript, React Native (Expo), Tailwind CSS, Redux Toolkit, TanStack Query, streaming chat UIs

Infra & Delivery: Docker & docker-compose, Gunicorn, Nginx, DigitalOcean, Vercel, GitHub Actions CI/CD, SonarQube, PyTest & Vitest, versioned releases

EDUCATION

Bachelor of Business Information Technology, Mount Kenya University

Information Technology, Software Development, and Artificial Intelligence.

CERTIFICATIONS

- Backend Engineer Certification
- AI Engineer Certification

REFERENCES

Available upon request.